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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP(HONOURS) EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCMAT100 - A GATEWAY TO MATHEMATICS

Scheme of Valuation

Duration: 2 Hours Maximum Marks: 70

Part A

Answer any 5 questions. Each question carries 2 Marks

Maximum marks from the section is 10

1. $\frac{dy}{dx} = \frac{5+12x^2}{-6}$ (2 Marks)

2. Define x = length, y = width, z = height.

   Write $x + (2y + 2z) = 108$ (1 Mark)

   Volume $V = x.y.z.$ ie, $V(y,z) = (108 - 2y - 2z).y.z$ (1 Mark)

3. $3x^2 - 3$ (2 Marks)

4. Evaluate the integral and obtain the answer $\frac{-1}{2}$ (2 Marks)

5. For solving $z = 5$ (1 Mark)

   For solving $y = 3 \,\&\, x = 1$ (1 Mark)

6. For getting the echelon form $\begin{bmatrix} 1 & 2 & 6 \\ 0 & -5 & -10 \end{bmatrix}$ (2 Marks)

   For getting $x = 2, y = 2$ (1 Mark)

7.

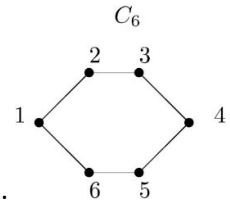
v1

/ \

e1 e2

/ \

v2---e3---v3 (2 Marks)

8. 

(2 Marks)

Part B

Answer any 5 questions. Each question carries 6 Marks

Maximum marks from this section is 30

9. $f_x = 2x - 2, f_y = -2y + 4$ (1 Mark)

The critical point is (1,2) (2 Marks)

Apply the second derivative test and obtain discriminant D = -4 (3 Marks)

Concluding (1,2) is a saddle point. (1 mark)

10.Writing Chain rule $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}.\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}.\frac{\partial y}{\partial u}$ (1 Mark)

$\frac{\partial z}{\partial x} = 4e^x \ln y$ (1 mark)

$\frac{\partial z}{\partial y} = \frac{4e^x}{y}$ (1 Mark)

$\frac{\partial x}{\partial u} = \frac{1}{u}$ (1 Mark)

$\frac{\partial y}{\partial u} = 1$ (1 Mark)

$\frac{\partial z}{\partial u} = 2\sqrt{2}.$ (2 Marks)

11. $\int_1^5 \left[x^3 - x^2y + y^2x\right]_{x=4}^{x=2} dy$ (2 Marks)

$= [56y - 6y^2 + \frac{2}{3}y^3]_{y=1}^{y=5}$ (2 Marks)

Getting the answer $\frac{388}{3}$ (2 Marks)

12. (a) For definition (2 Marks)

For writing the integral $\int_{-1}^{2} x^2 dx$ (1 Mark)

(b) For finding the length of each sub intervals (1 Mark)

For finding mesh (2 Marks)

13. For writing the matrix form A and find its rank (2 Marks)

For writing the augmented matrix $\tilde{A}$ and find its rank (2 Marks)

Rank (A) ≠ Rank ($\tilde{A}$) (1 Mark)

The system is inconsistent and hence has no solution (1 Mark)

14. For writing the matrix form (3 Marks)

For writing the augmented matrix (3 Marks)

15. A 2-regular graph is a graph in which every vertex has degree 2.

Example: Cycle graphs. (Number of vertices should be less than or equal to 5)

Each example carries 2 marks (6 Marks)

16. A 2-regular graph is a graph in which every vertex has degree 2.

Example: Cycle graphs. Each example carries 2 marks (6 Marks)

Part C

Answer any 3 questions. Each question carries 10 marks

Maximum marks of this section is 30

17. $f_x = \frac{1}{x+y}$, $f_y = \frac{1}{x+y}$ (2 Marks)

$f_{xx} = f_{yy} = \frac{-1}{(x+y)^2}$ (4 Marks)

$f_{xy} = f_{yx} = \frac{-1}{(x+y)^2}$ (4 Marks)

18. For writing the formula (2 Marks)

(a) $V = \int_3^5 . \int_1^2 (2x + y)\, dy\, dx = \int_3^5 2x + \frac{3}{2}\, dx = 19$ (4 Marks)

(b) $V = \int_0^2 . \int_0^3 x^2 dy\, dx = \int_0^2 3x^2 dx = 8$ (4 marks)

19. For statement (4 Marks)

Finding the antiderivative $F(t) = 2\sqrt{t} - 2t\sqrt{t}$ (2 Marks)

Applying the Fundamental Theorem $\int_1^4 \frac{1}{\sqrt{t}} - 3\sqrt{t})\, dt = 2\sqrt{t} - 2t\sqrt{t}\ ]_{t=1}^{t=4}$ (2 Marks)

Obtaining the answer = -12 (2 Marks)

20. For explanation (3 Marks)

For writing the matrix form A (2 marks)

For writing augmented matrix $\tilde{A}$ (2 Marks)

Obtain Rank (A) = Rank ($\tilde{A}$) = 3 and hence the system of equations is consistent (4 marks)

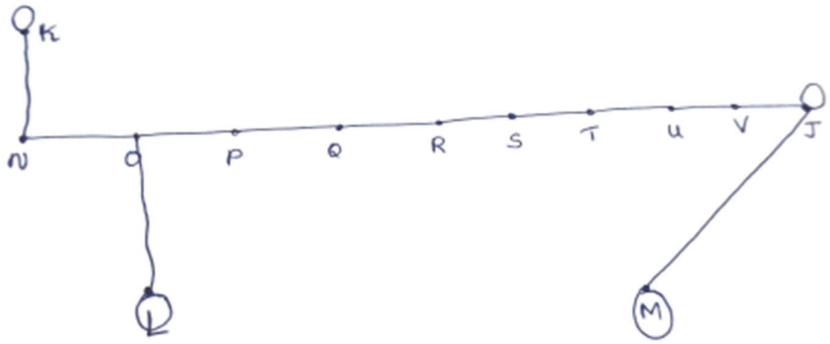
Obtain z = $\frac{-1}{4}$, y = $\frac{-61}{20}$, x = $\frac{12}{5}$ (3 Marks)

21. (a)Definition (3 Marks)

Degree sequence of P_5 = (2,2,2,1,1) (4 Marks)

(b) Statement (3 Marks)

22 a)



(8 marks)

b) Loops: JJ, KK, LL, MM (2 marks)

N.B Suitable stage marks may be fixed by the examiner for alternate correct answers.